Reusable component

DESIGN DOCUMENT + POST-DEPLOY RUNBOOK

FS Inspection Data Update Platform event publisher

Publishes the existing org platform event FS_InspectionDataUpdate__e when a Field Services Service Appointment changes status or scheduled start/end.

Org	AKLCDEVSB4
Status	Deployed
Deploy Id	0Af8r00000WnGE6CAN
Tests	FS_SaStatusEventHandlerTest 12 / 12 passing
Event	FS_InspectionDataUpdate__e (existing; not overwritten)
Event type	InspectionStatusUpdate (single type)
Source API	61.0
Classification	Confidential

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1. Purpose and tracker interpretation

When a Field Services inspection appointment changes **status** or **scheduled start / scheduled end**, Salesforce publishes one High Volume platform event so middleware can update downstream systems.

There is a single message type. Cancellation, completion, dispatch, travel, on-site, and a reschedule of SchedStartTime / SchedEndTime all use **InspectionStatusUpdate**. The JSON body always uses the same FS_InspectionStatusPayloadDTO shape. Properties with no value are omitted.

Tracker interpretation

The tracker item is not “only portal-created completions.” It is:

> *Do not publish for a Service Appointment that is a backfill appointment created for a manual booking created in CRM.*

Those backfills are stamped with Is Manual Inspection (FS_IsManualInspection__c = true) on Work Order, Work Order Line Item, and Service Appointment. The publisher evaluates the **Service Appointment** flag and skips the record.

2. Goals and non-goals

Goals

- Publish FS_InspectionDataUpdate__e after Field Services appointment status changes (including Canceled and unscheduled None).
- Also publish when SchedStartTime or SchedEndTime changes, even if status is unchanged.
- Always set header EventType__c to InspectionStatusUpdate.
- Map the full inspection payload; omit null JSON properties.
- Ignore appointments with FS_IsManualInspection__c = true.
- Ignore appointments that are not record type Field_Services.
- Sit beside existing Alert Center and FSL triggers; do not duplicate parent-lookup fill.

Non-goals

- Do not create additional platform event objects.
- Do not publish on insert (after update only).
- Do not publish solely because arrival window, due date, actuals, or FS_* date stamps changed.
- Do not publish Assigned Resource-only DML (no Service Appointment field change).
- Do not stamp FS_IsManualInspection__c from this package. CRM / booking automation must set it on WO, WOLI, and SA.
- Do not subscribe to the event in this package.

3. Architecture and deployed members

Salesforce cannot register a trigger on one record type. FS_SaStatusEventTrigger fires for every Service Appointment after update. Apex then filters.

```
ServiceAppointment after update
        |
        v
FS_SaStatusEventTrigger
        |  new MetadataTriggerHandler().run()
        v
Trigger Actions Framework
  sObject_Trigger_Setting.ServiceAppointment
  Trigger_Action.FS_SaStatusEventAfterUpdate
        |
        v
FS_SaStatusEventHandler.afterUpdate
        |-- keep Field_Services only
        |-- drop FS_IsManualInspection__c = true
        |-- require Default_Setting.Is_Active__c
        |-- require Status change OR SchedStartTime/SchedEndTime change
        v
SOQL (WITH SECURITY_ENFORCED, or SYSTEM_MODE for background users)
        v
FS_InspectionStatusEventMapper
        v
FS_InspectionDataUpdate__e
  EventType__c = InspectionStatusUpdate
  Payload__c   = JSON.serialize(DTO, suppress nulls)
```

Deployed members

Layer	Member	Role
Trigger	FS_SaStatusEventTrigger	TAF runner, after update only
Handler	FS_SaStatusEventHandler	Filters, qualifies, queries, publishes
Mapper	FS_InspectionStatusEventMapper	Header + DTO mapping
DTO	FS_InspectionStatusPayloadDTO	JSON names = Salesforce schema
Settings	FS_InspectionStatusEventSettings	Reads Default_Setting CMDT
Event	FS_InspectionDataUpdate__e	Existing org event. Not redeployed.
CMDT	Trigger_Action.FS_SaStatusEventAfterUpdate	Binds handler to ServiceAppointment
CMDT	sObject_Trigger_Setting.ServiceAppointment	TAF sObject setting
Perm set	FS_InspectionDataUpdate_Event_Publisher	Create event + read inspection fields

Parent Work Order lookups (FS_Work_Order__c, FS_Work_Order_Line_Item__c) are filled by Alert Center (FS_ServiceAppointmentTriggerHandler), not by this handler. The deploy left those Alert Center fields, Priority__c, and record type Field_Services unchanged.

4. Publication rules

An event is published when **all** of the following are true.

- CMDT FS_Inspection_Status_Event_Setting__mdt / Default_Setting has Is_Active__c true (packaged default is active if the record is missing).
- Record type is Field_Services. If that record type does not exist, nothing publishes.
- FS_IsManualInspection__c is not true on the Service Appointment.
- The update changed Status (any Field Services status, including None and Canceled), and/or SchedStartTime, and/or SchedEndTime.

If status and scheduled times change in the same DML, one event is published. The payload contains the new status and the new schedule.

Explicit non-publish cases

Situation	Result
Record type is not Field_Services	Skip
FS_IsManualInspection__c = true	Skip

Situation	Result
Default Setting inactive	Skip
Only ArrivalWindowStartTime / ArrivalWindowEndTime changed	Skip
Only DueDate, EarliestStartTime, actuals, or FS_* date stamps changed	Skip
Assigned Resource created/updated with no SA field change	Skip
Before insert / after insert	No trigger context in this package

Field Services statuses: None, Scheduled, Dispatched, Travel, OnSite, Completed, Canceled. There is no In Progress or Cannot Complete. Spreadsheet Onsite maps to OnSite.

5. Manual inspection flag

Object	API name	Label	Type	Default
WorkOrder	FS_IsManualInspection__c	Is Manual Inspection	Checkbox	false
WorkOrderLineItem	FS_IsManualInspection__c	Is Manual Inspection	Checkbox	false
ServiceAppointment	FS_IsManualInspection__c	Is Manual Inspection	Checkbox	false

All three are packaged so CRM can stamp the same meaning on the booking hierarchy. **Publication reads only the Service Appointment field.** Security classification is Confidential.

The process that creates a manual CRM booking and its Field Service backfill appointment must set FS_IsManualInspection__c = true on the Work Order, Work Order Line Item, and Service Appointment. If the SA flag is left false, this publisher will emit events. Portal / API booked inspections stay false (default) and continue to publish.

6. Platform event contract

Object FS_InspectionDataUpdate__e. High Volume. Publish After Commit. This object already existed in AKLCDEVSB4 and was not overwritten by the deploy.

Event field	Source	Notes
EventType__c	CMDT Event_Type__c	Always InspectionStatusUpdate
BookingReferenceNumber__c	WorkOrder.FS_Booking_Reference__c	Via FS_Work_Order__r
InspectionId__c	ServiceAppointment.FS_ExternalRefId__c	
WorkOrderId__c	ServiceAppointment.FS_Work_Order__c	
WorkOrderLineItemId__c	ServiceAppointment.FS_Work_Order_Line_Item__c	
ServiceAppointmentId__c	ServiceAppointment.Id	
Payload__c	JSON from FS_InspectionStatusPayloadDTO	Null properties omitted

Middleware should key off EventType__c = InspectionStatusUpdate and read payload status (and the rest of the body) rather than expecting InspectionCancellation or InspectionUpdate.

7. Payload (FS_InspectionStatusPayloadDTO)

Serialization: JSON.serialize(payload, true). A property is present only when it has a value. Nested assignedCrewMembers is omitted when there are no assigned resources (including unscheduled None). eventType inside the JSON matches the header.

Apex is case-insensitive, so the DTO cannot expose both userName and username. Display name is lastModifiedBy.name; login is lastModifiedBy.username.

Payload key	Salesforce source
eventType	Header event type

Payload key	Salesforce source
status	ServiceAppointment.Status
appointmentNumber	AppointmentNumber
priority	Priority__c (formula from parent WOLI / WO)
bookingReferenceNumber	WorkOrder.FS_Booking_Reference__c
inspectionId	FS_ExternalRefId__c
bcoNumber	BCO_Detail__c.FS_BCO_Number__c
workOrderLineItemNumber	WorkOrderLineItem.LineItemNumber
workTypeName	WorkType.Name
bcoComplexity	BCO_Detail__c.FS_BCO_Complexity__c
arrivalWindowStartTime / EndTime	ArrivalWindowStartTime / EndTime
schedStartTime / schedEndTime	SchedStartTime / SchedEndTime
dispatchedDate	FS_Dispatched_Date__c
dueDate	DueDate
earliestStartTime	EarliestStartTime
cancelledDate	FS_Cancelled_Date__c
travelDate / onSiteDate / completedDate	FS_Travel_Date__c / FS_OnSite_Date__c / FS_Completed_Date__c
estimatedTravelTimeMinutes	First AssignedResource.EstimatedTravelTime (minutes)
estimatedTravelTimeSeconds	Minutes x 60
inJeopardy / inJeopardyReason	FSL__InJeopardy__c / FSL__InJeopardyReason__c
inspectionStatus	FS_Inspection_Status__c
cancelledReason	FS_Cancelled_Reason__c
bookingType	FS_Booking_Type__c
isConfirmed	FS_Is_Confirmed__c
parentWork.*	Work Order / WOLI ids and numbers
lastModifiedBy.name / username / employeeNumber	LastModifiedBy Name, Username, EmployeeNumber
assignedCrewMembers[]	AssignedResource related list

Nested parentWork

Key	Source
workOrderId	FS_Work_Order__c
workOrderNumber	FS_Work_Order__r.WorkOrderNumber
bookingReferenceNumber	FS_Work_Order__r.FS_Booking_Reference__c
workOrderLineItemId	FS_Work_Order_Line_Item__c
workOrderLineItemNumber	FS_Work_Order_Line_Item__r.LineItemNumber
workOrderLineItemExternalRefId	FS_Work_Order_Line_Item__r.FS_ExternalRefId__c

Nested assignedCrewMembers[]

Key	Source
recordId	AssignedResource.Id

Key	Source
serviceCrewId / serviceCrewName	ServiceCrew
serviceResourceId / serviceResourceName	ServiceResource
serviceResourceEmployeeNumber	ServiceResource.RelatedRecord.EmployeeNumber
estimatedTravelTimeMinutes / Seconds	EstimatedTravelTime
schedStartTime / schedEndTime	Parent appointment schedule

When status is the configured unscheduled value (None), assigned resources are not added to the payload. Travel time in Salesforce is minutes. Field Service may recalculate EstimatedTravelTime on assign.

Schema gaps (not mapped)

Spreadsheet idea	Org reality	Payload
WorkOrder FS_Sequence__c	Field does not exist	workOrderLineItemNumber
ServiceResource FS_ExternalRefId__c	Field does not exist	related User EmployeeNumber
AssignedResource Glick-key	No field	Omitted

8. Class design

FS_SaStatusEventHandler

Implements TriggerAction.AfterUpdate.

Method	Responsibility
afterUpdate	Cast TAF lists, keep publishable appointments, call handleTrigger
filterPublishableAppointments / isPublishableAppointment	Field_Services and not manual
resolveEventType	InspectionStatusUpdate if status or sched start/end changed; else null
queryAppointments	Payload SOQL; WITH SECURITY_ENFORCED or AccessLevel.SYSTEM_MODE
excludeManualInspections	Second pass on queried rows
publishEvents	EventBus.publish; tests capture publishedEventsForTest

Background users CloudIntegrationUser and AutomatedProcess cannot be granted FLS. Those user types query and publish in system mode. Org automation “FS Service Appointment After Create” requires FS_BCO_Detail__c (bcoDetailId). That is an org flow, not this package; tests populate BCO Detail.

FS_InspectionStatusEventMapper

Builds FS_InspectionDataUpdate__e and FS_InspectionStatusPayloadDTO. Null DTO properties are left null so JSON suppress-nulls drops them.

FS_InspectionStatusEventSettings

Reads Default_Setting. Blank Event_Type__c falls back to InspectionStatusUpdate. Blank Unscheduled_Status__c falls back to None. Tests inject settingsOverride or forceMissingSettingForTest.

9. Configuration

Custom metadata type FS_Inspection_Status_Event_Setting__mdt, record Default Setting (deployed):

Field	Packaged default	Purpose
Is_Active__c	true	Master publish switch

Field	Packaged default	Purpose
Event_Type__c	InspectionStatusUpdate	Header and payload eventType
Unscheduled_Status__c	None	Status that omits assigned crew from the payload
Scope_To_Field_Services_Record_Type__c	true	Layout compatibility; Apex always requires Field_Services
Description__c	(text)	Admin notes

Removed from this design (do not use):

- Separate cancellation / dates event types
- Portal-creator filter on Completed
- Skip-on-enter status list
- Tracked-status allowlist (every Field Services status change publishes)

10. Security

Permission set FS_InspectionDataUpdate_Event_Publisher grants:

- Apex access to handler, mapper, settings, DTO
- Create + read FS_InspectionDataUpdate__e and its fields
- Read inspection fields used in the publisher query
- Read/edit FS_IsManualInspection__c on WorkOrder, WorkOrderLineItem, ServiceAppointment
- Record type visibility ServiceAppointment.Field_Services

SOQL uses WITH SECURITY_ENFORCED for interactive users. EventBus.publish runs after isCreateable() (or system-mode bypass for background users). Custom fields created in this package use securityClassification Confidential.

11. Automated tests

FS_SaStatusEventHandlerTest calls afterUpdate explicitly so tests do not depend on TAF CMDT already being in the org. Setup creates Operating Hours, Territory, Work Order, WOLI, BCO Detail, and a Field Services Service Appointment.

Test	Expectation
Parent lookups on insert	Alert Center filled FS_Work_Order__c / WOLI / Priority
Enter Scheduled	One InspectionStatusUpdate
Canceled	One InspectionStatusUpdate with WO / WOLI / booking reference
Non-Field_Services	No event
FS_IsManualInspection__c = true	No event
Change SchedStartTime / SchedEndTime	One InspectionStatusUpdate with schedule in payload
Assigned resource travel mapping	Payload includes assignedCrewMembers and travel minutes
Arrival window only	resolveEventType returns null
Unschedule to None	One InspectionStatusUpdate
Inactive CMDT	No event
Missing / blank CMDT	Defaults to active + InspectionStatusUpdate + None

12. Post-deploy manual steps (AKLCDEVSB4)

Complete these in the org. Metadata deploy does **not** assign permission sets, add fields to page layouts, stamp CRM backfills, or configure subscribers.

12.1 Assign the publisher permission set

- Setup → Users → Permission Sets → FS_InspectionDataUpdate_Event_Publisher.
- Manage Assignments. Assign to Field Service dispatchers who change appointment status or schedule.
- Assign to field resources / mobile users who move appointments through Travel, OnSite, Completed, Canceled.
- Assign to integration users that update Field Services appointments and must publish the event.
- Do not expect to assign this set to CloudIntegrationUser / AutomatedProcess. Those user types cannot take Field Service FLS; the handler already queries and publishes in system mode for them.

Without this assignment, interactive users fail WITH SECURITY_ENFORCED on the payload query or cannot create FS_InspectionDataUpdate__e.

12.2 Field-level security for other profiles

If some users update Service Appointments using profile FLS instead of this permission set:

- Grant Read on ServiceAppointment.FS_IsManualInspection__c (required for the publisher query).
- Grant Read on the other inspection fields already listed on the permission set (BCO, booking reference, FS date stamps, jeopardy, parent lookups, Status).
- Grant Create on FS_InspectionDataUpdate__e for anyone who should publish.
- Prefer assigning the permission set rather than editing many profiles.

12.3 Add Is Manual Inspection to page layouts

The checkbox was deployed but is not automatically on page layouts.

- Work Order layout used by CRM booking: add Is Manual Inspection.
- Work Order Line Item layout used by CRM booking: add Is Manual Inspection.
- Service Appointment Field Services layout: add Is Manual Inspection. Keep it editable for the CRM / booking process that stamps backfills. Optionally read-only for dispatchers if they should not uncheck it.
- Until the field is on the layout, admins can still set it via Inspector, Data Loader, or Apex.

12.4 Confirm custom metadata Default Setting

- Setup → Custom Metadata Types → FS Inspection Status Event Setting → Manage Records → Default Setting.
- Is Active = true.
- Event Type = InspectionStatusUpdate.
- Unscheduled Status = None.
- Leave it active unless you need an emergency kill switch.

12.5 Confirm Trigger Actions Framework wiring

- sObject Trigger Setting record ServiceAppointment: Bypass Execution = false; Object API Name = ServiceAppointment.
- Trigger Action record FS_SaStatusEventAfterUpdate: After Update bound to the ServiceAppointment sObject setting; Apex Class Name = FS_SaStatusEventHandler; Bypass Execution = false; Order = 1 (or an agreed order next to other future SA actions).
- Confirm only one unpackaged Service Appointment trigger calls new MetadataTriggerHandler().run(). A second TAF runner on the same object will double-publish.
- Leave FSL-managed TR001_Service_* and Alert Center FS_ServiceAppointmentTrigger in place. This package does not replace them.

12.6 Subscriber / middleware access

- Grant the subscriber integration user Read on FS_InspectionDataUpdate__e (subscribe). Create remains on the publisher set.
- Point middleware at FS_InspectionDataUpdate__e, not at three separate event objects.
- Branch only on EventType__c = InspectionStatusUpdate. Status, cancellation, and reschedule share that value; use payload status and the date fields to distinguish.
- Treat omitted JSON keys as null. Do not expect InspectionCancellation or InspectionUpdate as EventType__c.
- Map lastModifiedBy.name (display name) and lastModifiedBy.username (login).

12.7 CRM manual-booking backfill process (required)

The publisher does not infer “manual” from CreatedBy or UserType. When CRM creates a manual booking and a Field Service backfill appointment, set FS_IsManualInspection__c = true on Work Order, Work Order Line Item, and Service Appointment.

If the Service Appointment flag stays false, events will publish. Optional data fix: if historical backfill appointments already exist, set the checkbox true on those SAs (and parents) so they stop emitting.

12.8 Smoke test in AKLCDEVSB4

Use a Field Services appointment with BCO Detail populated (org after-create automation requires bcoDetailId).

Step	Action	Expected
1	Unscheduled Field Services SA to status Scheduled (set sched start/end)	One event, EventType__c = InspectionStatusUpdate, status = Scheduled, Inspection Id and booking reference populated
2	Change SchedStartTime / SchedEndTime only	One event, same event type, new times in payload
3	Change arrival window only	No event
4	Status Canceled	One event, status = Canceled, cancelledDate if FS_Cancelled_Date__c is set
5	Check Is Manual Inspection, then change status	No event
6	Non-Field_Services appointment status change	No event
7	Dispatcher vs integration user	Event still publishes (system mode for background users)

12.9 Sign-off checklist

- [ ] Permission set assigned to dispatcher, field, and publisher integration users
- [ ] Is Manual Inspection on WO, WOLI, and Field Services SA layouts
- [ ] Default Setting active, Event Type InspectionStatusUpdate
- [ ] TAF ServiceAppointment setting and FS_SaStatusEventAfterUpdate not bypassed
- [ ] No second MetadataTriggerHandler on Service Appointment
- [ ] Subscriber can read the platform event
- [ ] Middleware updated for a single event type and schema-named payload keys
- [ ] CRM stamps FS_IsManualInspection__c on manual backfills
- [ ] Smoke tests 1-6 passed

13. What was not deployed

- FS_InspectionDataUpdate__e object and fields (already in the org; source-tracking conflict).
- Alert Center FS_Work_Order__c, FS_Work_Order_Line_Item__c, Priority__c.
- Record type ServiceAppointment.Field_Services (already in the org).
- TAF object definitions (Trigger_Action__mdt, sObject_Trigger_Setting__mdt) — only the Service Appointment records were deployed.
- Alert Center and FSL managed trigger bodies.